



38 million pieces of trash

An uninhabited island in the middle of the Pacific is covered with estimated 38 million pieces of trash. <http://dgit.in/SeaTrash>



Rare dinosaur fossil found

Zuul crurivastator, named after a Ghostbusters demon, was recently found with preserved skeleton and skin. <http://dgit.in/ZuulDino>

Sometimes, size does matter

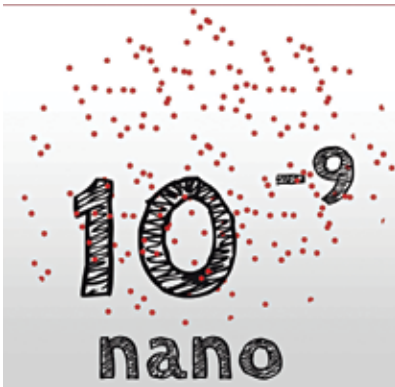
From diamond clothing to chemical bond microscopes – here are the latest innovations from the field of nanotechnology

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anotechnology is going to reshape our future. It's already been doing that for some years now but

with the world at large finally awakening to the possibilities of literally harnessing particles a little bigger than 10^{-9} meters to do many, many things, the growth is going to be explosive. Here are some cutting-edge developments in the field



That's way too many zeros 0.0

01 ATOM FORGES

What if we could craft an item by literally forging it atom by atom? Ever since IBM scientists spelt out their company's name using 35 Xenon Atoms way back in 1989 using a Scanning Tunnelling Microscope, people have been trying to find a viable way of arranging atoms in 3D to make objects (The Scanning Tunnelling Mi-

croscope can only affect the top layer of atoms so you can only make 2D Objects) Scientists have also known for long that while electron microscopy provides quite reliable and accurate magnification, it has an annoying tendency to nudge an atom or two out of its place in the lattice due to the stream of electrons it emits. But with the advancements in heavy computing and deep learning, it is now possible to use this effect for good. A particular variety of Electron Microscope known as Scanning Transmission Electron Microscope or STEM is used to focus the beam of electrons on a particular atom in a material and nudge it to a new place. At the same time, the magnification provided by the STEM is used to analyse the new structure of the material with the atom moved and a computer decides to either move the same atom some more or nudge a different atom. If you prefer a more scientific explanation, here's some further reading : <http://dgit.in/AtmFrge>. This cycle repeats over and over again until all the required atoms form a new lattice and thereby a completely new material, which can offer up exciting avenues for developing new items that have specific purposes.

02 MOVE OVER BLURAY, GLASS DISCS ARE THE NEW BEST THING

There used to be a time when people used to actually struggle to fill 700 MB CD ROMs. Now with the rapid adoption of faster bandwidths and people clamouring for higher and higher quality, even 128 gig BluRay discs are fast becoming too much of a constraint. Enter the 'glass disc'. It's actually very similar to a normal CD-ROM. In a CD, there are several narrow concentric rings or 'tracks' expanding outward from the centre. When you write something to the CD, a laser beam burns a hole or a 'pit' 1mm deep to represent a 1 in binary. The places where there is not pit is called 'land' and this represents 0. So when you finish writing the CD and pop it into a reader, a laser beam is focussed on each track in turn and it is reflected back differently based on whether there is a pit or land and the reader, therefore, 'reads' the data as the CD rotates. Now, for a glass disc, they use a special laser to burn in data in nanoparticles and there are FIVE unique states to mark it instead of just two. And since we are dealing with orders of nanometers rather than millimetres